

Ans 1

```
#include<conio.h>
#include<conio.h>
void main()
{
int i,a[10];
clrscr();
printf("Enter elements of array");
for(i=1;i<=10;i++)
scanf("\n%d",&a[i]);
printf("\nThe elements you entered are: ");
for(i=1;i<=10;i++)
printf("\n%d",a[i]);
getch();
}
```

Ans 2

```
#include<stdio.h>
#include<conio.h>
void main()
{
int sum=0,max,i,avg=0,a[10];
clrscr();
printf("Enter total no of students: ");
scanf("%d",&max);
printf("Enter marks of students");
for(i=1;i<=max;i++)
{
scanf("%d",&a[i]);
sum=sum+a[i];
}
avg=sum/max;
printf("\nAverage of marks is: %d",avg);
getch();
}
```

Ans 3

```
#include<stdio.h>
#include<conio.h>
void main()
{
int limit,i,a[10],min,max;
clrscr();
printf("Enter total no of values: ");
scanf("%d",&limit);
printf("Enter the values: ");
for(i=1;i<=limit;i++)
scanf("%d",&a[i]);
min=max=a[1];
for(i=1;i<=limit;i++)
{
if(a[i]>max)
max=a[i];
}
```

```

if(a[i]<min)
min=a[i];
}
printf("\n Min: %d \n Max: %d",min,max);
getch();
}

```

Ans 4

```

#include<stdio.h>
#include<conio.h>
void main()
{
int limit,i,a[10],j,temp;
clrscr();
printf("Enter total no of values: ");
scanf("%d",&limit);
printf("Enter the values: ");
for(i=1;i<=limit;i++)
scanf("%d",&a[i]);
for(i=1;i<=limit;i++)
{
for(j=1;j<=limit-i;j++)
{
if(a[j]>a[j+1])
{
temp=a[j];
a[j]=a[j+1];
a[j+1]=temp;
}
}
}
printf("\n\nThe sorted list is: \n");
for(i=1;i<=limit;i++)
printf("\n%d",a[i]);
getch();
}

```

Ans 5

```

#include<stdio.h>
#include<conio.h>
void main()
{
int limit1,limit2,limit3,i,j,a[10],b[10],c[20];
clrscr();
printf("Enter total no of elements in first array: ");
scanf("%d",&limit1);
printf("Enter the values: ");
for(i=0;i<limit1;i++)
scanf("%d",&a[i]);

printf("Enter total no of elements in second array: ");
scanf("%d",&limit2);
printf("Enter the values: ");
for(i=0;i<limit2;i++)

```

```

scanf("%d",&b[i]);

limit3=limit1+limit2;
for(i=0;i<limit1;i++)
c[i]=a[i];

for(i=0,j=limit1;j<limit3;j++,i++)
c[j]=b[i];

printf("\n\n The merged array is: \n");
for(i=0;i<limit3;i++)
printf("%d\n",c[i]);
getch();
}

```

If no of elements in both the arrays are same then i.e. a[5],b[5],c[10]

```

for(i=0;i<10;i++)
{
if(i<5)
c[i]=a[i];
else
c[i]=b[5-i]
}

```

Ans 6

```

#include<stdio.h>
#include<conio.h>
#include<process.h>
void main()
{
int a[2][2],b[2][2],c[2][2],i,j,k,choice;
char cont='n';
clrscr();
do
{
printf("\n\t\tMENU\n\t\t_____");
printf("\n1. Addition\n2.Subtraction\n3.Multiplication\n4.Division\n5.Exit");
printf("\n\nEnter your choice: ");
scanf("%d",&choice);

if(choice==1||choice==2||choice==3||choice==4)
{
printf("\nEnter elements for first matrix -> ");
for(i=1;i<=2;i++)
{
for(j=1;j<=2;j++)
{
scanf("%d",&a[i][j]);
}
}
}

printf("\nEnter elements for second matrix -> ");
for(i=1;i<=2;i++)

```

```

{
for(j=1;j<=2;j++)
{
scanf("%d",&b[i][j]);
}
}
}
switch(choice)
{
case 1:
    for(i=1;i<=2;i++)
    for(j=1;j<=2;j++)
    c[i][j]=a[i][j]+b[i][j];

    printf("\nAddition of matrices is ->\n");

    for(i=1;i<=2;i++)
    {
    for(j=1;j<=2;j++)
    {
    printf("%d\t",c[i][j]);
    }
    printf("\n");
    }
    break;
case 2:    for(i=1;i<=2;i++)
    for(j=1;j<=2;j++)
    c[i][j]=a[i][j]-b[i][j];

    printf("\nSubtraction of matrices is ->\n");

    for(i=1;i<=2;i++)
    {
    for(j=1;j<=2;j++)
    {
    printf("%d\t",c[i][j]);
    }
    printf("\n");
    }
    break;
case 3:

    for(i=1;i<=2;i++)
    {
    for(j=1;j<=2;j++)
    {
    c[i][j]=0;
    }
    }

    for(i=1;i<=2;i++)
    {
    for(j=1;j<=2;j++)
    {

```

```

        for(k=1;k<=2;k++)
        {
            c[i][j]=c[i][j]+a[i][k]*b[k][j];
        }
    }
}

printf("\nMultiplication of matrices is ->\n");
for(i=1;i<=2;i++)
{
    for(j=1;j<=2;j++)
    {
        printf("%d\t",c[i][j]);
    }
    printf("\n");
}
break;
case 4: for(i=1;i<=2;i++)
        for(j=1;j<=2;j++)
            c[i][j]=a[i][j]/b[i][j];

printf("\nDivisionn of matrices is ->\n");

for(i=1;i<=2;i++)
{
    for(j=1;j<=2;j++)
    {
        printf("%d\t",c[i][j]);
    }
    printf("\n");
}
break;
case 5: exit(0);
default:printf("\nEnter correct choice: ");
}
printf("\nDo you wish to continue(Y/N)? ");
cont=getch();
clrscr();
}while((cont=='y')||(choice!=5));
getch();
}

```

Ans 7

```

#include<stdio.h>
#include<conio.h>
#include<process.h>
void main()
{
    int a[3][3],i,j,row_sum,col_sum;
    clrscr();

    printf("\nEnter elements of the matrix -> ");
    for(i=0;i<3;i++)
    {

```

```

for(j=0;j<3;j++)
{
scanf("%d",&a[i][j]);

}
}
printf("\n\nThe matrix you have entered is ->\n");
for(i=0;i<3;i++)
{
for(j=0;j<3;j++)
{
printf("%d\t",a[i][j]);
}
printf("\n");
}

printf("\n\nSum of rows is ->\n");
for(i=0;i<3;i++)
{
row_sum=0;
for(j=0;j<3;j++)
{
row_sum=row_sum+a[i][j];
}
printf("\n\nSum of row: %d is -> %d",i,row_sum);
}

printf("\n\n\nSum of columns is ->\n");
for(j=0;j<3;j++)
{
col_sum=0;
for(i=0;i<3;i++)
{
col_sum=col_sum+a[i][j];
}
printf("\n\nSum of column: %d is -> %d",j,col_sum);
}
getch();
}

```

Ans 8

```

#include<stdio.h>
#include<conio.h>
#include<process.h>

void main()
{
int a[3][3],b[3][3],i,j,row_sum,col_sum;
clrscr();

printf("\nEnter elements of the matrix -> ");
for(i=1;i<=3;i++)
{
for(j=1;j<=3;j++)

```

```

{
scanf("%d",&a[i][j]);
}
}

printf("\n\nThe matrix you have entered is ->\n");
for(i=1;i<=3;i++)
{
for(j=1;j<=3;j++)
{
printf("%d\t",a[i][j]);
}
printf("\n");
}

for(i=1;i<=3;i++)
{
for(j=1;j<=3;j++)
{
b[i][j]=a[j][i];
}
}
printf("\n");
}

printf("\n\nThe transpose of matrix is ->\n");
for(i=1;i<=3;i++)
{
for(j=1;j<=3;j++)
{
printf("%d\t",b[i][j]);
}
}
printf("\n");
}
getch();
}

```

Ans 9 i)

```

#include<stdio.h>
#include<conio.h>
void main()
{
int a[10],elem,i,n,flag;
clrscr();
printf("Enter the no of elements -> ");
scanf("%d",&n);
printf("\nEnter the elements -> ");
for(i=1;i<=n;i++)
scanf("%d",&a[i]);
printf("\n\nEnter the elements to be searched -> ");
scanf("%d",&elem);

for(i=1;i<=n;i++)
{

```

```

if(elem==a[i])
{
flag=1;
break;
}
else
flag=0;
}
if(flag==1)
printf("\nElement is found at position -> %d",i);
else
printf("\nElement not found!!!");
getch();
}

```

Ans 9 ii)

```

#include<stdio.h>
#include<conio.h>
void main()
{
int a[10],i,elem,mid,start,end,flag;
clrscr();
printf("Enter the no of elements -> ");
scanf("%d",&end);
printf("\nEnter the elements -> ");
for(i=1;i<=end;i++)
scanf("%d",&a[i]);
printf("\n\nEnter the elements to be searched -> ");
scanf("%d",&elem);
start=1;
mid=(end+1)/2;
if(elem>mid)
start=mid;
else
end=mid;
while(start<=end)
{
if(elem==a[start])
{
flag=1;
break;
}
else
{
flag=0;
}
start++;
}
if(flag==1)
printf("\nElement is found at position -> %d",start);
else
printf("\nElement not found!!!");
getch();
}

```


Ans 9

```
#include<stdio.h>
#include<conio.h>

void sqr()
{
    int i;
    printf("\nSqure of numbers between 1 & 10 is->\n\n");
    for(i=1;i<=10;i++)
        printf("\n\t%d ",(i*i));
}

void cube()
{
    int i;
    printf("\n\nCube of numbers between 1 & 10 is->\n\n");
    for(i=1;i<=10;i++)
        printf("\n\t%d ",(i*i*i));
}

void main()
{
    clrscr();
    sqr();
    cube();
    getch();
}
```

Ans 10

```
#include<stdio.h>
#include<conio.h>
int fun_sum(int);
void main()
{
    int max,ans;
    clrscr();
    printf("Enter the max number till which you want to add: ");
    scanf("%d",&max);
    ans=fun_sum(max);
    printf("\nSum of numbers from 1 to %d is: %d",max,ans);
    getch();
}

int fun_sum(n)
{
    int i,s=0;
    for(i=1;i<=n;i++)
        s=s+i;
    return(s);
}
```

Ans 11

```
#include<stdio.h>
#include<conio.h>
int check(int);

void main()
{
    int ans,n;
    clrscr();
    printf("Enter a number -> ");
    scanf("%d",&n);
    ans=check(n);
    if(ans==1)
        printf("Number is even");
    else
        printf("Number is odd");
    getch();
}

int check(num)
{
    if(num%2==0)
        return 1;
    else
        return 0;
}

/*
void check();

void main()
{
    clrscr();
    check();
    getch();
}

void check()
{
    int num;
    printf("Enter a number -> ");
    scanf("%d",&num);
    if(num%2==0)
        printf("\n Number is even");
    else
        printf("Number is odd");
}
*/
```

Ans 12

```
#include<stdio.h>
```

```

#include<conio.h>
void sum(int,int);
void main()
{
int start,end;
clrscr();
printf("Enter a starting number -> ");
scanf("%d",&start);
printf("\nEnter a ending number -> ");
scanf("%d",&end);
sum(start,end);
getch();
}

void sum(s,e)
{
int i,sum_even=0,sum_odd=0;
for(i=s;i<=e;i++)
{
if(i%2==0)
sum_even=sum_even+i;
else
sum_odd=sum_odd+i;
}
printf("\nSum of even numbers is -> %d",sum_even);
printf("\nSum of odd numbers is -> %d",sum_odd);
}

```

Ans 13 i

```

#include<stdio.h>
#include<conio.h>
int fact(int num);
void main()
{
int ans,n;
clrscr();
printf("Enter a number -> ");
scanf("%d",&n);
ans=fact(n);
printf("\nFactorial of the %d is -> %d",n,ans);
getch();
}

```

```

int fact(num)
{
int f=1;
while(num>0)
{
f=f*num;
num--;
}

```

```

/* OR
for(;num>0;num--)
f=f*num;

```

```

        */
return f;
}

```

Ans 13 ii

```

#include<stdio.h>
#include<conio.h>
int fact(int num);
void main()
{
    int ans,n;
    clrscr();
    printf("Enter a number -> ");
    scanf("%d",&n);
    ans=fact(n);
    printf("\nFactorial of the %d is -> %d",n,ans);
    getch();
}

```

```

int fact(num)
{
    if(num==0)
        return 1;
    else
        return(num*fact(num-1));
}

```

Ans 14 i

```

#include<stdio.h>
#include<conio.h>
void fibo(int);
void main()
{
    int ans,n;
    clrscr();
    printf("Enter the number till which you want fibonacci series -> ");
    scanf("%d",&n);
    fibo(n);
    getch();
}

```

```

void fibo(max)
{
    int a=0,b=1,c,count=2;
    if(max<=0)
        printf("No series can be generated");
    else if(max==1)
        printf("%d",a);
    else
    {
        printf("%5d%5d",a,b);
        while(count<max)

```

```

{
c=a+b;
printf("%5d",c);
a=b;
b=c;
count++;
}
}
}

```

Ans 14 ii

```

#include<stdio.h>
#include<conio.h>
int fibo(int);
void main()
{
int i=0,n,c;
clrscr();
printf("Enter the number till which you want fibonacci series -> ");
scanf("%d",&n);
for(c=1;c<=n;c++)
{
printf("%5d",fibo(i));
i++;
}
getch();
}

int fibo(max)
{
if(max<=1)
return max;
else
return(fibo(max-1)+fibo(max-2));
}

```

Ans 15 i

```

#include<stdio.h>
#include<conio.h>

void exchange(int,int);

void main()
{
int a,b;
clrscr();
printf("\nEnter first number -> ");
scanf("%d",&a);
printf("\nEnter second number -> ");
scanf("%d",&b);
printf("\n\nIn main Original values are:");

```

```

printf("\n a=%d \n b=%d",a,b);
exchange(a,b);
printf("\n\n After comming back to main values are: \n");
printf("\n a=%d \n b=%d",a,b);
getch();
}

```

```

void exchange(int x,int y)
{
int temp;
temp=x;
x=y;
y=temp;
printf("\n\nIn function interchanged values are:");
printf("\n a=%d \n b=%d",x,y);
}

```

Ans 15 ii

```

#include<stdio.h>
#include<conio.h>

```

```

void exchange(int *,int *);

```

```

void main()
{
int a,b;
clrscr();
printf("\nEnter first number -> ");
scanf("%d",&a);
printf("\nEnter second number -> ");
scanf("%d",&b);
printf("\n\nIn main Original values are:");
printf("\n a=%d \n b=%d",a,b);
exchange(&a,&b);
printf("\n\n After comming back to main values are: \n");
printf("\n a=%d \n b=%d",a,b);
getch();
}

```

```

void exchange(int *x,int *y)
{
int temp;
temp=*x;
*x=*y;
*y=temp;
printf("\n\nIn function interchanged values are:");
printf("\n a=%d \n b=%d",*x,*y);
}

```

Assgn 2 Prac (call by val/ref)

```

#include<stdio.h>
#include<conio.h>

/*void inc(int);

void main()
{
int a,b;
clrscr();
printf("\nEnter a number -> ");
scanf("%d",&a);
printf("\n\nIn main Original value is:");
printf("\n a=%d",a);
inc(a);
printf("\n\n After comming back to main value is: \n");
printf("\n a=%d",a);
getch();
}

void inc(int x)
{
x=x+10;
printf("\n\nIn function incremented value is:");
printf("\n a=%d",x);
} */

```

```

void inc(int *);

void main()
{
int a,b;
clrscr();
printf("\nEnter a number -> ");
scanf("%d",&a);
printf("\n\nIn main Original value is:");
printf("\n a=%d",a);
inc(&a);
printf("\n\n After comming back to main value is: \n");
printf("\n a=%d",a);
getch();
}

void inc(int *x)
{
*x=*x+10;
printf("\n\nIn function incremented value is:");
printf("\n a=%d",*x);
}

```